

CAMERA SHELL DESIGN: DRAG ESTIMATES

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Scope. This is an inviscid, two-dimensional wedge estimate for a camera-shell compression face. It is useful for selecting a slender transition angle, but it is not a flight qualification analysis: skin friction, base drag, three-dimensional interference, thermal loads, structural loads, and stability are omitted.

1. Governing result

For a shell face of area S inclined by angle θ to the rocket/fuselage axis, with approximately uniform post-shock pressure p_2 , the axial wave-drag component is

$$D_w = (p_2 - p_\infty) S \sin \theta \quad (1)$$

where p_∞ is the freestream static pressure. The pressure p_2 is obtained from the attached oblique-shock solution:

$$\tan \theta = 2 \cot \beta \frac{M_\infty^2 \sin^2 \beta - 1}{M_\infty^2 (\gamma + \cos 2\beta) + 2}, \quad (2)$$

$$\frac{p_2}{p_\infty} = 1 + \frac{2\gamma}{\gamma + 1} (M_\infty^2 \sin^2 \beta - 1). \quad (3)$$

For the illustrative case $M_\infty = 3$, $\gamma = 1.4$, and $\theta = 10^\circ$, the weak attached-shock solution is $\beta \simeq 27.4^\circ$, $p_2/p_\infty \simeq 2.05$, and $C_p = (p_2 - p_\infty)/q_\infty \simeq 0.167$.

2. Shock and drag schematic

The left panel shows the sharp-wedge idealization used by Eqs. (2) and (3). The right panel shows what changes when the shell nose is rounded: the shock generally detaches, and its stand-off distance depends on nose radius and three-dimensional geometry rather than on the wedge relation alone.

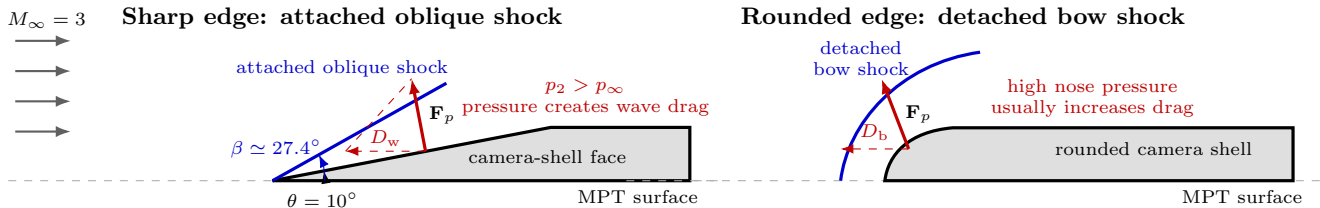


Figure 1: Schematic of the shock system and pressure-drag mechanism. The shock locations are illustrative and are not to scale.

Freestream pressure versus altitude. For a standard atmosphere, the entire 5,000–30,000 ft interval lies in the troposphere. With h_{ft} denoting geometric altitude in feet, $h_{\text{m}} = 0.3048h_{\text{ft}}$, $T_0 = 288.15$ K, $p_0 = 101325$ Pa, lapse rate $L = 0.0065$ K/m, $R = 287.05$ J/(kg K), and $g_0 = 9.80665$ m/s², using the NASA standard-atmosphere model [given here](#).

$$p_{\infty}(h_{\text{ft}}) = 101325 \left(1 - 6.8756 \times 10^{-6} h_{\text{ft}}\right)^{5.2559} \text{ Pa}, \quad 5000 \leq h_{\text{ft}} \leq 30000. \quad (4)$$

At $M_{\infty} = 3$ and $\gamma = 1.4$, $q_{\infty} = \frac{1}{2}\gamma M_{\infty}^2 p_{\infty} = 6.3p_{\infty}$. Representative values are:

altitude (ft)	5,000	10,000	15,000	20,000	25,000	30,000
p_{∞} (kPa)	84.3	69.7	57.2	46.6	37.6	30.1
q_{∞} at $M = 3$ (kPa)	531.1	439.0	360.2	293.3	236.9	189.6

Example drag density at 5,000 ft. For $h = 5000$ ft, $\theta = 10^\circ$, and the weak-shock result $p_2/p_{\infty} = 2.0545$,

$$\frac{D_w}{S} = (p_2 - p_{\infty}) \sin \theta = [(2.0545 - 1)(84.3 \text{ kPa})] \sin 10^\circ \simeq 15.4 \text{ kPa} = 15.4 \text{ kN/m}^2 \quad (\simeq 2.24 \text{ psi}). \quad (5)$$

Last year's design. For the previous design, $S = 9 \times 10^{-4} \text{ m}^2$ and $\theta = 8^\circ$. At 5,000 ft and $M_{\infty} = 3$, the weak-shock solution gives $\beta \simeq 25.61^\circ$ and $p_2/p_{\infty} \simeq 1.7953$. Thus the estimated full axial wave drag on the shell face is

$$D_w = [(1.7953 - 1)(84.3 \text{ kPa})] (9 \times 10^{-4} \text{ m}^2) \sin 8^\circ \simeq 8.40 \text{ N} \quad (\simeq 1.89 \text{ lbf}). \quad (6)$$

For comparison, CFD results for last year's design gave $D_{\text{CFD}} \simeq 7 \pm 3 \text{ N}$.

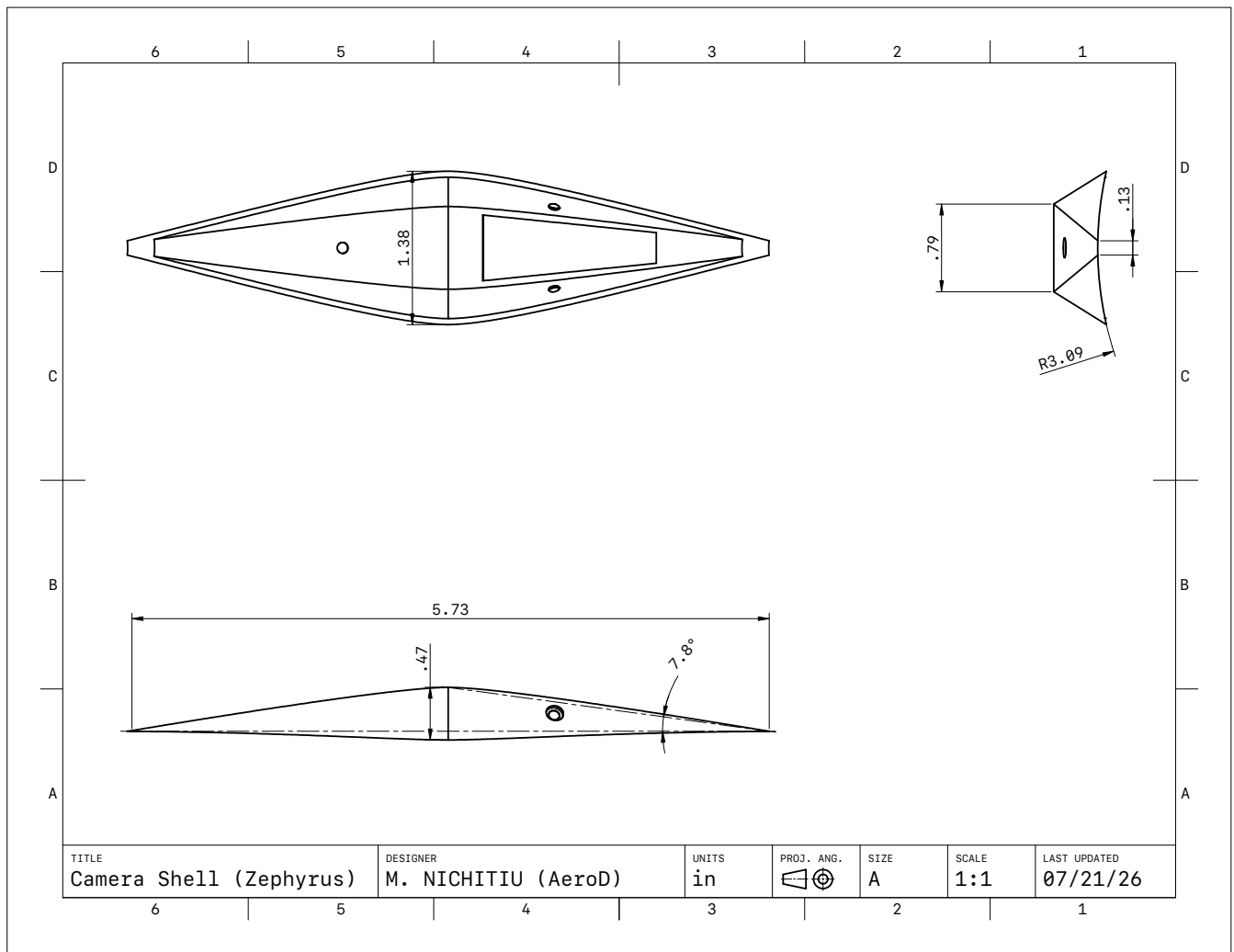


Figure 2: Engineering Sketch of the Camera Shell for *Project Zephyrus*.

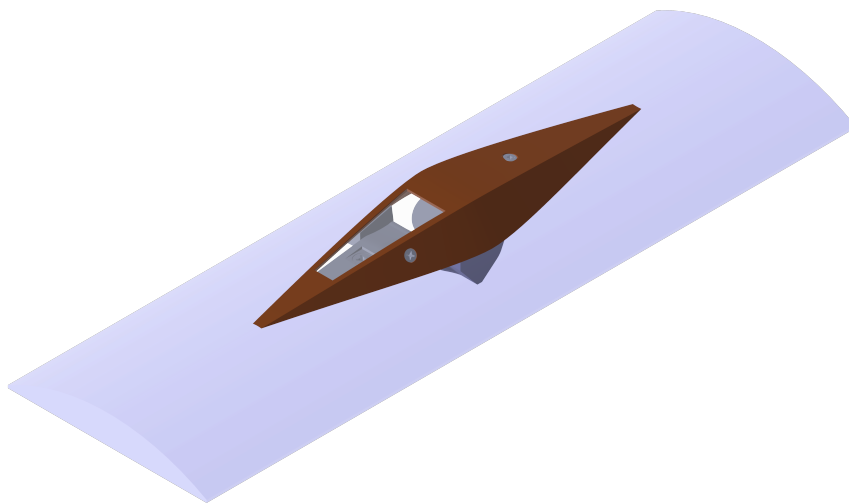


Figure 3: Render of the Camera Shell for *Project Zephyrus*.

3. Derivation of the drag relation

Let $\hat{\mathbf{x}}$ point downstream, in the direction of the external flow in the body-fixed frame, and let $\mathbf{n}$ be the outward unit normal of the shell. The pressure traction on a surface element dS is

$$d\mathbf{F} = -p \mathbf{n} dS. \quad (7)$$

The uniform freestream pressure produces no net force on a closed body, so it is useful to work with the pressure difference $\Delta p = p - p_\infty$. The drag magnitude, defined positive opposite the flow direction, is therefore

$$D = -\mathbf{F} \cdot \hat{\mathbf{x}} = \int_S (p - p_\infty) n_x dS, \quad n_x = \mathbf{n} \cdot \hat{\mathbf{x}}. \quad (8)$$

For a sharp, uniform compression face inclined at θ to the axis, the face normal has an axial projection $n_x = \sin \theta$. Oblique-shock theory treats the surface pressure as uniform at p_2 . Substituting both facts into Eq. (8) gives

$$D_w = \int_S (p_2 - p_\infty) \sin \theta dS = (p_2 - p_\infty) S \sin \theta, \quad (9)$$

which is Eq. (1).

Using the freestream dynamic pressure

$$q_\infty = \frac{1}{2} \gamma p_\infty M_\infty^2, \quad (10)$$

the corresponding pressure-drag coefficient is

$$C_{D,w} = \frac{D_w}{q_\infty S_{\text{ref}}} = C_p \frac{S}{S_{\text{ref}}} \sin \theta, \quad C_p = \frac{p_2 - p_\infty}{q_\infty}. \quad (11)$$

The pressure p_2 follows by treating the component of Mach number normal to the shock as a normal-shock problem:

$$M_{n,\infty} = M_\infty \sin \beta, \quad \frac{p_2}{p_\infty} = 1 + \frac{2\gamma}{\gamma + 1} (M_{n,\infty}^2 - 1), \quad (12)$$

while the flow-turning requirement gives the θ - β - M relation in Eq. (2). The weak-shock root is normally selected when it exists.

4. Angle selection

For small angles, the pressure rise across a weak oblique shock is approximately proportional to θ , while $\sin \theta \approx \theta$. Consequently,

$$D_w \propto \theta^2 \quad (\theta \ll 1). \quad (13)$$

Therefore oblique-shock theory alone gives no nonzero optimum angle: if the shell height is fixed and its axial transition length can grow, drag is reduced by making the shell more slender. For a prescribed height h and transition length L ,

$$\theta = \arctan\left(\frac{h}{L}\right). \quad (14)$$

Use the smallest practical angle with a smooth, tangent-continuous fairing. Rounded or blunt shells require CFD or detached-bow-shock analysis; Δx in Fig. 1 is illustrative rather than determined by the sharp-wedge equations.

Sources: NASA Glenn, [oblique-shock relations](#); NASA Glenn, [Mach 2.5 wedge validation](#); NASA Glenn, [standard-atmosphere model](#).